The Schrieffer Wolff Transformation

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1 Introduction

Systems in many-body physics frequently involve a weak perturbation ϵV to a decoupled system H_0 which mixes these systems. It is common to call the uncoupled system the "bare" system and the inclusion of the weak perturbation the "dressed" system, with bare/dressed eigenstates and energy levels respectively. Examples of systems where this situation arises include Rydberg atoms, with the Hamiltonian [1]

$$H = -\Delta \sum_i n_i + \frac{C}{2} \sum_{i,j} \frac{n_i n_j}{r_{ij}^6} + \frac{\Omega}{2} \sum_i \sigma_i^x \tag{1}$$

This describes an array of neutral atoms at positions r_i which interact via the $\sim \frac{1}{r^6}$ Van der Waals force, where Ω is the laser power and Δ is the laser detuning from resonance. If Ω is small compared to $\frac{C}{r^6}$, then this leads to the Rydberg blockade effect, where the subspace of states with no adjacent excitations is separated from the rest of the spectrum by a gap and weakly coupled by Ω. Another relevant situation is the interaction between a multilevel atom and a resonant cavity [2]:

$$H = \sum_j \omega_j |j\rangle\langle j| + \omega_r a^\dagger a + \sum_{ij} g_{ij} |i\rangle\langle j| \, a^\dagger + \text{h.c.} \tag{2}$$

Although formally the coupling is unbounded, for small enough excitation numbers these couplings can be treated as small with respect to the bare system (see Appendix B of [2]). Lastly, the original use of Schrieffer and Wolff was to relate the Kondo and Anderson model Hamiltonians [4]. The Anderson model Hamiltonian for a single localized d orbital is

$$H = \sum_{ks} \epsilon_k n_{ks} + \sum_s \epsilon_d n_{ds} + U n_{d\uparrow} n_{d\downarrow} + \sum_{ks} V_{kd} c_{ks}^\dagger c_{ds} + \text{h.c.} \tag{3}$$

where ϵ_k and ϵ_d are the single-particle energies of the localized and conduction orbitals respectively. The potential V mixes these states, and U is the Coulomb repulsion between two d-orbital electrons. The idea of the Schrieffer-Wolff transformation is to construct a Hamiltonian H_{eff} which acts on the low-energy subspace of H_0 but reproduces the eigenvalues of $H = H_0 + \epsilon V$, obtaining an effective interaction between the bare states.

2 Direct rotations

The essence is to define a canonical rotation between two subspaces, which will be the low-energy subspaces of the unperturbed and perturbed Hamiltonian respectively. This canonical rotation is called a "direct rotation," which we will define in the following section.

Def 1. *Given two subspaces P_0, P_1 of the same dimension, the unitary U closest to the identity that maps $P_0 \to P_1$ is called the direct rotation relating the two subspaces.*

Def 2. *Given a subspace P_0, we define a reflection R_0 about that subspace via $R_0 = 2\pi_0 - 1$, where π_0 is a projector onto P_0.*

Lemma 1. $\|\pi_1 - \pi_0\| < 1$ *iff no vector in P_0 is orthogonal to P_1 and vice versa.*

Proof. For the forward direction, suppose the contrary. Then WLOG $v \in P_1 \perp P_0$, so $\|(\pi_1 - \pi_0)v\| = 1$, which is a contradiction. For the reverse direction, again suppose the contrary. Then the maximum eigenvalue of $\pi_1 - \pi_0$ is ± 1, so there exists v such that $(\pi_1 - \pi_0)v = \pm v$. This is possible only if $\pi_0 v = 0$ or $\pi_1 v = 0$, which completes the proof. □

Lemma 2. *We may block-diagonalize R_1 and R_0 with blocks of size less than or equal to two on the diagonal.*

Proof. TBD. ☐

Lemma 3. *Suppose that P_0 and P_1 are subspaces and $\|\pi_1 - \pi_0\| < 1$. Then $\mathsf{T} = \pi_1 - \pi_0$ satisfies $\|\mathsf{T}\| = \|\pi_0\pi_1^\perp\| = \|\pi_0^\perp\pi_1\|$.*

Proof. Considering again the block-diagonal form, we need only prove the claim for each block. We have $\mathsf{T}^\dagger\mathsf{T} = \pi_0 + \pi_1 - \pi_0\pi_1 - \pi_1\pi_0$, and thus $\pi_0^\perp\mathsf{T}^\dagger\mathsf{T}\pi_0 = 0$, so $\mathsf{T}^\dagger\mathsf{T}$ is block-diagonal with respect to π_0. Thus the largest eigenvalue of $\mathsf{T}^\dagger\mathsf{T}$ is the largest eigenvalue of either $\pi_0\mathsf{T}^\dagger\mathsf{T}\pi_0 = \pi_0\pi_1^\perp\pi_0$ or $\pi_0^\perp\mathsf{T}^\dagger\mathsf{T}\pi_0^\perp = \pi_0^\perp\pi_1\pi_0^\perp$, which is either $\|\pi_0\pi_1^\perp\|$ or $\|\pi_0^\perp\pi_1\|$. Since $\mathsf{P}_0, \mathsf{P}_0^\perp, \mathsf{P}_1, \mathsf{P}_1^\perp$ are all one-dimensional subspaces, it is clear that $\|\pi_0^\perp\pi_1\| = \|\pi_0\pi_1^\perp\|$. ☐

Prop 1. *If $\|\pi_1 - \pi_0\| < 1$, then the unitary U_{01} given by $\sqrt{\mathsf{R}_1\mathsf{R}_0}$ maps P_0 to P_1, where this is well-defined when the branch cut of the square root is taken along $\mathbb{R}^{<0}$.*

Proof. Since P_0 and P_1 may not contain any subspaces orthogonal to the other, we focus on the 2×2 blocks. Then P_0 is spanned by $|0\rangle$ and P_1 is spanned by $\cos(\theta)|0\rangle + \sin(\theta)|1\rangle$, where $|0\rangle, |1\rangle$ are normalized kets spanning the block in question. Explicitly, we have $\mathsf{R}_0 = \sigma^z$ and

$$\begin{aligned}
\mathsf{R}_1 &= 2(\cos(\theta)|0\rangle + \sin(\theta)|1\rangle)(\cos(\theta)\langle 0| + \sin(\theta)\langle 1|) - 1 && (4)\\
&= 2\cos^2(\theta)|0\rangle\langle 0| + 2\sin^2(\theta)|1\rangle\langle 1| + 2\sin(\theta)\cos(\theta)\sigma^x - 1 && (5)\\
&= 2|1\rangle\langle 1| + 2\cos^2(\theta)\sigma^z + \sin(2\theta)\sigma^x - 1 && (6)\\
&= 2|1\rangle\langle 1| + \cos(2\theta)\sigma^z + \sigma^z + \sin(2\theta)\sigma^x - 1 && (7)\\
&= \cos(2\theta)\sigma^z + \sin(2\theta)\sigma^x && (8)
\end{aligned}$$

Therefore

$$\mathsf{R}_1\mathsf{R}_0 = \exp(-2i\theta\sigma^y) \tag{9}$$

This shows that $\mathsf{U}_{01} = \exp(-i\theta\sigma^y)$, and therefore performs the desired rotation. Furthermore, since $\theta < \pi/2$, the square root is well-defined. ☐

Cor 1. *U_{01} satisfies $\mathsf{U}_{01}\mathsf{P}_1\mathsf{U}_{01}^\dagger = \mathsf{P}_0$.*

It is more useful later on to work directly with the generator, whose properties we address in the following lemma:

Lemma 4. *The generator S of U_{01} is block-off-diagonal with respect to P_0, satisfies $\exp(\mathsf{S})\mathsf{P}_1\exp(-\mathsf{S}) = \mathsf{P}_0$ and $\|\mathsf{S}\| < \frac{\pi}{2}$. Furthermore, S satisfying these properties is unique.*

Proof. We will not prove uniqueness. The properties follow from the previous proof. ☐

With uniqueness, we can formally identify the construction above with the initial definition of a direct rotation.

3 The Schrieffer-Wolff transformation

The Schrieffer-Wolff transformation is a second-order perturbation theory technique to block-diagonalize a Hamiltonian into low-energy and high-energy subspaces. Given a bare Hamiltonian H_0 and eigenspaces $\mathsf{V}_<, \mathsf{V}_>$, let $\mathsf{H} = \mathsf{H}_0 + \epsilon\mathsf{V}$ be the system Hamiltonian, where V does not preserve the decomposition $\mathsf{V}_< \oplus \mathsf{V}_>$. Then we wish to find a unitary transformation U, called the Schrieffer-Wolff transformation, such that $UHU^\dagger$ reproduces the spectrum of H when restricted to $V_<$. This will be the direct rotation relating the low-energy subspaces of H and H_0. We say that H_0 has a spectral gap Δ if there exists an interval I_0 such that any eigenvalue of H_0 in I_0 is separated from the rest of the spectrum by at least Δ. Define P_0 to be the space spanned by eiegenvalues of H_0 within I_0. Then let I be the interval obtained by thickening I_0 by $\Delta/2$. Define P_1 to be the subspace spanned by eigenvectors of H within I. Intuitively, the perturbation ϵV can shift the spectrum by at most ϵ, so for $\epsilon < \Delta/2$, all of the eigenvalues of H_0 within I_0 will correspond to eigenvalues of H within I. This is expressed in the following lemma:

Lemma 5. *Suppose* $\|V\| = 1$ *and* $\epsilon < \Delta/2$. *Then* $\|\pi_1 - \pi_0\| \leqslant 2\epsilon/\Delta < 1$

Proof. Put $T = \pi_1 - \pi_0$. By a proposition in the previous section, we have $\|T\| = \|\pi_0\pi_1^\perp\|$. Then we observe that

$$(\pi_0 H_0 \pi_0)(\pi_0\pi_1^\perp) - (\pi_0\pi_1^\perp)(\pi_1^\perp H \pi_1^\perp) = \pi_0(H_0 - H)\pi_1^\perp = -\epsilon\pi_0 V \pi_1^\perp \tag{10}$$

We freely subtract a constant from H_0 so that I_0 is centered at zero. Since the spectrum of $(\pi_0 H_0 \pi_0)$ is separated from the spectrum of $(\pi_1^\perp H \pi_1^\perp)$ by at least $\Delta/2$, by the lemma that follows we have $\|\pi_0\pi_1^\perp\| \leqslant \frac{2\epsilon}{\Delta}$. □

Lemma 6. *Let* A, B *be normal operators such that* $\mathrm{spec}(A) \subseteq D_\rho$ *and* $\mathrm{spec}(B) \subseteq D^c_{\rho+\delta}$. *Then the solution of the equation* $BX - XA = Y$ *(called a Sylvester equation) satisfies* $\|X\| \leqslant \frac{1}{\delta}\|Y\|$.

Proof. First, observe that

$$X = \sum_{n=0}^{\infty} B^{-n-1} Y A^n \tag{11}$$

Satisfies the given equation. Furthermore, the operator norm is submultiplicative, so

$$\|X\| \leqslant \sum_{n=0}^{\infty} \|B^{-1}\|^{n+1} \|Y\| \|B\|^n \tag{12}$$

$$\leqslant \frac{\|Y\|}{\rho+\delta} \sum_{n=0}^{\infty} \left(\frac{\rho}{\rho+\delta}\right)^n \tag{13}$$

$$= \frac{1}{\delta}\|Y\| \tag{14}$$

□

Def 3. *The direct rotation relating* P_0 *and* P_1 *(as defined at the beginning of the section) is called the Schrieffer-Wolff transformation, and the operator* $H_{\mathit{eff}} = \pi_0 U_{01} H U_{01}^\dagger \pi_0$ *is called an effective Hamiltonian.*

4 Perturbation Theory

Without diagonalizing both Hamiltonians, it is difficult to construct the transformation directly. Instead, we can use the fact (proved in the last section) that S is off-diagonal in π_0 and H is block-diagonal in π_0.

Def 4. *Define the super-operators* $\mathcal{O}$ *and* $\mathcal{D}$ *as projecting onto the off-diagonal and diagonal blocks respectively with respect to* π_0. *Let* $\mathcal{L}$ *be a superoperator defined by* $\mathcal{L}(|i\rangle\langle j|) = \mathcal{O}(|i\rangle\langle j|)/(E_i - E_j)$, *where* $H_0 |i\rangle = E_i |i\rangle$.

Prop 2. *If* X *is an operator, then*

$$\mathcal{L}([H_0, X]) = [H_0, \mathcal{L}(X)] = \mathcal{O}(X) \tag{15}$$

Proof.

$$\mathcal{L}([H_0, X]) = \sum_{i,j} \frac{\langle i| \mathcal{O}([H_0, X]) |j\rangle}{E_i - E_j} |i\rangle\langle j| \tag{16}$$

$$= \sum_{i,j} \frac{\langle i| H_0\mathcal{O}(X) - \mathcal{O}(X)H_0) |j\rangle}{E_i - E_j} |i\rangle\langle j| \tag{17}$$

$$= \sum_{i,j} \frac{(E_i - E_j)\langle i| \mathcal{O}(X) |j\rangle}{E_i - E_j} |i\rangle\langle j| = \sum_{i,j} \langle i| \mathcal{O}(X) |j\rangle |i\rangle\langle j| = \mathcal{O}(X) \tag{18}$$

□

Lemma 7. *S satisfies the equation* $S = \epsilon\left[(\mathcal{D}(V)) + \hat{S}\coth\left(\hat{S}\right)\mathcal{O}(V)\right]$

Proof. Write $\hat{S}$ for the superoperator action of S by the commutator. Then we may write $H_{\text{eff}} = UHU^\dagger = \exp(S)H\exp(-S) = \exp\left(\hat{S}\right)H$, which is block-diagonal with respect to π_0 as shown previously. Then

$$\mathcal{O}(\exp\left(\hat{S}\right)H) = \sinh\left(\hat{S}\right)\mathcal{D}(H) + \cosh\left(\hat{S}\right)\mathcal{O}(H) = \sinh\left(\hat{S}\right)\mathcal{D}(H) + \epsilon\cosh\left(\hat{S}\right)\mathcal{O}(V) = 0 \tag{19}$$

Since S is block-off-diagonal,

$$S = \mathcal{O}(S) = \mathcal{L}([H_0, S]) = -\mathcal{L}(\hat{S}H_0) \tag{20}$$

Assuming that ϵ is sufficiently small such that $\sinh\left(\hat{S}\right)$ is invertible, we have $\mathcal{D}(H) = H_0 + \epsilon\mathcal{D}(V) = -\epsilon\coth\left(\hat{S}\right)\mathcal{O}(V)$ from before, so

$$\mathcal{L}(\hat{S}H_0) = -\epsilon\mathcal{L}(\hat{S}\mathcal{D}(V) - \hat{S}\coth\left(\hat{S}\right)\mathcal{O}(V)) \tag{21}$$

□

Lemma 8. *If* $S = \sum_{n=0}^{\infty}\epsilon^n S_n$ *is the Taylor series expansion of* S, *then* S_n *satisfies* $S_{n+1} = \mathcal{L}\hat{S}_n\mathcal{D}(V) + \sum_{j=1}^{\infty} a_{2j}\mathcal{L}(\hat{S}^{2j}_{n-1}\mathcal{O}(V))$, *where we define the symbol*

$$\hat{S}^k_m = \sum_{\substack{n_1,\ldots,n_k=1 \\ n_1+\cdots+n_k=m}} \hat{S}_{n_1}\ldots\hat{S}_{n_k} \tag{22}$$

and a_m *are the coefficients in the Taylor series expansion of* $x\coth(x)$.

Proof. We solve the equation from the previous lemma with $S = \sum_{n=0}^{\infty}\epsilon^n S_n$:

$$\sum_{n=0}^{\infty}\epsilon^n S_n = \sum_{n=0}^{\infty}\epsilon^{n+1}\mathcal{L}\hat{S}_n\mathcal{D}(V) + \epsilon\mathcal{L}\sum_{m=0}^{\infty}a_m S^m\mathcal{O}(V) \tag{23}$$

$$= \sum_{n=0}^{\infty}\epsilon^{n+1}\mathcal{L}\hat{S}_n\mathcal{D}(V) + \epsilon\mathcal{L}\sum_{m=0}^{\infty}a_m\left(\sum_n \epsilon^n\hat{S}_n\right)^m\mathcal{O}(V) \tag{24}$$

$$= \sum_{n=0}^{\infty}\epsilon^{n+1}\mathcal{L}\left[\hat{S}_n\mathcal{D}(V) + \sum_{m=0}^{\infty}a_m\hat{S}^n_k\mathcal{O}(V)\right] \tag{25}$$

Equating powers of ϵ on each side gives the desired result. □

Lemma 9. *The effective Hamiltonian is given by*

$$H_{eff} = P_0 H P_0 + P_0 \sum_{n=2}^{\infty} \epsilon_n \left[\sum_{j=1}^{\infty} b_{2j-1} \hat{S}_{n-1}^{2j-1} \mathcal{O}(V) \right] P_0 \tag{26}$$

Where b_m *are the coefficients in the Taylor series of* $\tanh(x/2)$.

Proof. Assuming $\cosh\left(\hat{S}\right)$ is invertible, which holds for sufficiently small S, we have $\tanh\left(\hat{S}\right)\mathcal{D}(H) = -\epsilon\mathcal{O}(V)$. Then the transformed Hamiltonian can be rewritten as

$$\exp\left(\hat{S}\right)H = \mathcal{D}(\exp\left(\hat{S}\right)H) = \cosh\left(\hat{S}\right)\mathcal{D}(H) + \sinh\left(\hat{S}\right)\mathcal{O}(H) \tag{27}$$

$$= \mathcal{D}(H) + \frac{\cosh\left(\hat{S}\right) - 1}{\tanh\left(\hat{S}\right)} \tanh\left(\hat{S}\right)\mathcal{D}(H) + \epsilon \sinh\left(\hat{S}\right)\mathcal{O}(V) \tag{28}$$

$$= \mathcal{D}(H) + \epsilon \left[\sinh\left(\hat{S}\right) - \frac{\cosh\left(\hat{S}\right) - 1}{\tanh\left(\hat{S}\right)} \right] \mathcal{O}(V) \tag{29}$$

$$= \mathcal{D}(H) + \epsilon \tanh\left(\hat{S}/2\right)\mathcal{O}(V) \tag{30}$$

Combining this with the series expression for S from the previous lemma gives the desired result. □

5 Locality bounds

For the remainder of this section, consider a bipartition of the system into subsystems A and B, and let O^A denote an operator that acts nontrivially only on subsystem A.

Lemma 10. *If* $\|\pi_1^A - \pi_0^A\| < 1$ *and* $\|\pi_1^B - \pi_0^B\| < 1$, *then this implies* $\|\pi_1^A \otimes \pi_1^B - \pi_0^A \otimes \pi_0^B\| < 1$.

Proof. By assumption, we have $\pi_1^A - \pi_0^A \leqslant (1-\alpha)I$ and $\pi_1^B - \pi_0^B \leqslant (1-\beta)I$ for $\alpha, \beta > 0$ (We interpret $M \geqslant N$ as saying that $M - N$ is positive semi-definite). Rearranging, we find $\pi_1^A\pi_0^A\pi_1^A \geqslant \alpha\pi_1^A$ and $\pi_1^B\pi_0^B\pi_1^B \geqslant \beta\pi_1^B$, which implies that

$$(\pi_1^A \otimes \pi_1^B)(\pi_0^A \otimes \pi_0^B)(\pi_1^A \otimes \pi_1^B) \geqslant \alpha\beta\pi_1^A \otimes \pi_1^B \tag{31}$$

where we use the fact that the tensor product preserve positive semi-definiteness. However proceeding by contradiction, if $\|\pi_1^A \otimes \pi_1^B - \pi_0^A \otimes \pi_0^B\| = 1$, then there is a vector ν such that $\pi_1^A \otimes \pi_1^B\nu = \nu$ (or likewise for 0) and $\pi_0^A \otimes \pi_0^B\nu = 0$. However applying the equation above to ν gives $0 \geqslant \alpha\beta$, which is a contradiction. □

Lemma 11.

$$(U^{AB})^\dagger(\pi_1^A \otimes \pi_1^B) = (U^A \otimes U^B)^\dagger(\pi_1^A \otimes \pi_1^B) \tag{32}$$

Proof. The above lemma implies that we can block-diagonalize the projectors π_1^A, π_0^A, and π_1^B, π_1^B, we only need to prove it for a 2×2 block. However, this follows quickly from the fact that each block is a projector onto a one-dimensional subspace, i.e. $(U^A)^\dagger \left|\psi_1^A\right\rangle = \left|\psi_0^A\right\rangle$ and $(U^B)^\dagger \left|\psi_1^B\right\rangle = \left|\psi_0^B\right\rangle$ implies that $(U^A \otimes U^B)^\dagger \left|\psi_1^A\right\rangle \otimes \left|\psi_1^B\right\rangle = \left|\psi_0^A\right\rangle \otimes \left|\psi_0^B\right\rangle$. □

Lemma 12. *Let* $H_0 = H_0^A + H_0^B$ *and* $V = V^A + V^B$. *Then*

$$H_{eff} = H_{eff}^A + H_{eff}^B \tag{33}$$

where H_{eff}^A *is the SW transformation obtained for* H_0^A *to* $H_0^A + \epsilon V^A$ *and similarly for* B.

Proof. Now we have

$$
\begin{aligned}
H_{eff}^{AB} &= (\pi_0^A \otimes \pi_0^B) U^{AB} H (U^{AB})^\dagger (\pi_0^A \otimes \pi_0^B) && (34)\\
&= (\pi_0^A \otimes \pi_0^B)(U^A \otimes U^B) H (U^A \otimes U^B)^\dagger (\pi_0^A \otimes \pi_0^B) && (35)\\
&= H_{eff}^A + H_{eff}^B && (36)
\end{aligned}
$$

□

Def 5. *Consider an interaction graph* (Λ, E)*. Expand* $V = \sum_{e \in E} \epsilon_e V^e$*, where* $e \in E$ *indexes the terms* ϵ_e *and* V^e*. Then we can expand*

$$H_{eff} = \sum_{n=0}^{\infty} \sum_{C \subset E}^{\infty} K_n^C \tag{37}$$

where $K_n^C \equiv \sum_{m_1+m_2+\cdots+m_n=n} \prod_{e_i \in C} \epsilon_{e_i}^{m_i} K_{m_1,\ldots,m_n}^C$ *for some operators* $K_{m_1,\ldots,m_n}^C$

Prop 3. K_n^C *acts only on* C *and* $K_n^C = 0$ *unless* C *is a connected set and* $|C| \leqslant n$*.*

Proof. Since K_n^C does not depend on ϵ_e for $e \notin C$, we can freely take these to be zero. Then the system factorizes into $\Lambda(C)$ and its compliment, so the additivity property shows that K_n^C only acts on $\Lambda(e) \in C$. For the second claim, suppose $C = C_1 \sqcap C_2$. Then in a similar method, we set $\epsilon_e = 0$ for any $e \notin C$, and the system factorizes into a product of C_1 and C_2. Thus $H_{eff} = H_{eff}^A + H_{eff}^B$, which an affords an expansion in which no term contains variables ϵ^e from both C_1 and C_2. The claim that $K_n^C = 0$ for $|C| > n$ is just by definition. □

Note that the above proof is very general, and only relies on the additivity property of the transformation.

References

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